

Detailed Solutions to Tutorial Sheet 4

Probability and Stochastic Processes

Solution to Q1

- The function is given as $F(x, y) = 1 - e^{-y}$ for $0 \leq x, y < \infty$, and 0 otherwise.
- For $F(x, y)$ to be a valid joint distribution function, the probability of any rectangular region must be non-negative.
- Consider a region defined by $x_1 < x_2$ and $y_1 < y_2$. The probability of this region is given by $P(x_1 < X \leq x_2, y_1 < Y \leq y_2) = F(x_2, y_2) - F(x_1, y_2) - F(x_2, y_1) + F(x_1, y_1)$.
- For $0 \leq x_1 < x_2$ and $0 \leq y_1 < y_2$, this yields $(1 - e^{-y_2}) - (1 - e^{-y_2}) - (1 - e^{-y_1}) + (1 - e^{-y_1}) = 0$. This is non-negative.
- For $x_1 < 0 \leq x_2$ and $0 \leq y_1 < y_2$, $F(x_1, \cdot) = 0$. The probability calculation gives $F(x_2, y_2) - 0 - F(x_2, y_1) + 0 = (1 - e^{-y_2}) - (1 - e^{-y_1}) = e^{-y_1} - e^{-y_2}$. Since $y_1 < y_2$, $e^{-y_1} > e^{-y_2}$, making this difference strictly positive.
- The asymptotic limits hold: $\lim_{x \rightarrow \infty, y \rightarrow \infty} F(x, y) = 1$, and limits as $x \rightarrow -\infty$ or $y \rightarrow -\infty$ are exactly 0.
- Furthermore, the function is right-continuous in both variables.
- Conclusion: Yes, it is the joint distribution function of a pair of random variables (specifically, X is deterministically 0, and Y is exponentially distributed).

Solution to Q2

- To show that (X, Y) is jointly continuous, we must demonstrate that the mixed partial derivative $f(x, y) = \frac{\partial^2 F}{\partial x \partial y}$ exists and integrates back to $F(x, y)$.
- First, compute the partial derivative with respect to x for $x \geq 0$: $\frac{\partial F}{\partial x} = e^{-x} \left(\frac{1}{2} + \frac{1}{\pi} \tan^{-1}(y) \right)$.
- Next, differentiate this result with respect to y : $f(x, y) = \frac{\partial^2 F}{\partial x \partial y} = e^{-x} \frac{1}{\pi(1+y^2)}$ for $x \geq 0$.
- For $x < 0$, $f(x, y) = 0$.
- Integrating $f(x, y)$ over the region $(-\infty, x] \times (-\infty, y]$ exactly yields $F(x, y)$.
- Since the cumulative distribution function can be recovered completely by integrating a non-negative density function over the domain, the pair (X, Y) is jointly continuous.

Solution to Q3

- The joint PMF must sum to 1 over all possible values: $\sum_{x=1}^3 \sum_{y=1}^3 cxy = 1$.
- Factoring out c , we get $c \left(\sum_{x=1}^3 x \right) \left(\sum_{y=1}^3 y \right) = 1$.

- Evaluating the sums gives $c(1+2+3)(1+2+3) = c(6)(6) = 36c = 1$. Thus, $c = \frac{1}{36}$.
- To find $\mathbb{P}(1 \leq X \leq 2, Y \leq 2)$, we sum the probabilities over the restricted range: $\sum_{x=1}^2 \sum_{y=1}^2 \frac{xy}{36} = \frac{(1+2)(1+2)}{36} = \frac{9}{36} = \frac{1}{4}$.
- To find $\mathbb{P}(Y = 3)$, we sum over all values of X while fixing $Y = 3$: $\sum_{x=1}^3 \frac{x(3)}{36} = \frac{3(1+2+3)}{36} = \frac{18}{36} = \frac{1}{2}$.

Solution to Q4

- **Part i:** Because X and Y are independent, the joint probability mass function factors into the marginals: $p_{(X,Y)}(x,y) = p_X(x)p_Y(y)$.
- Using $p_{(X,Y)}(4,2) = 0.18$ and $p_Y(2) = 0.6$, we can solve for $p_X(4)$: $p_X(4) = \frac{0.18}{0.6} = 0.3$.
- The total probability for X is 1, so $p_X(1) = 1 - p_X(3) - p_X(4) = 1 - 0.5 - 0.3 = 0.2$.
- The total probability for Y is 1, so $p_Y(1) = 1 - p_Y(2) = 1 - 0.6 = 0.4$.
- The joint distribution $p_{(X,Y)}(\cdot, \cdot)$ is formed by multiplying all pairwise combinations of p_X and p_Y .
- **Part ii:** We need to compute $\mathbb{P}(|X - Y| \geq 2)$. We first find all pairs (x,y) satisfying the condition.
- The valid pairs are $(3,1)$, $(4,1)$, and $(4,2)$.
- The required probability is $p_{(X,Y)}(3,1) + p_{(X,Y)}(4,1) + p_{(X,Y)}(4,2) = (0.5)(0.4) + (0.3)(0.4) + (0.3)(0.6)$.
- Calculating the terms gives $0.20 + 0.12 + 0.18 = 0.5$.

Solution to Q5

- **Part a:** First, establish the distribution of $Z = XY$. The possible values for X and Y are $\{-1, 1\}$, each with probability $\frac{1}{2}$.
- $Z = 1$ occurs when $(X = 1, Y = 1)$ or $(X = -1, Y = -1)$. Since X and Y are independent, $\mathbb{P}(Z = 1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$.
- Similarly, $Z = -1$ occurs for $(1, -1)$ and $(-1, 1)$, giving $\mathbb{P}(Z = -1) = \frac{1}{2}$.
- We check pairwise independence between X and Z . $\mathbb{P}(X = 1, Z = 1) = \mathbb{P}(X = 1, Y = 1) = \frac{1}{4}$. This equals $\mathbb{P}(X = 1)\mathbb{P}(Z = 1) = (\frac{1}{2})(\frac{1}{2}) = \frac{1}{4}$.
- By symmetry, all pairs $\{(X, Y), (X, Z), (Y, Z)\}$ are pairwise independent.
- **Part b:** For mutual independence, the joint probability of all three variables must equal the product of their marginals.
- Check the case $\mathbb{P}(X = 1, Y = 1, Z = -1)$. Since $Z = XY$, if $X = 1$ and $Y = 1$, then Z must be 1. Thus, $\mathbb{P}(X = 1, Y = 1, Z = -1) = 0$.
- However, $\mathbb{P}(X = 1)\mathbb{P}(Y = 1)\mathbb{P}(Z = -1) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} \neq 0$.
- Thus, X , Y , and Z are *not* mutually independent.

Solution to Q6

- Since X and Y are independent geometric variables, their joint probability is the product of their marginal probabilities.
- We want to find $\mathbb{P}(X + Y = z)$ for a fixed integer $z > 2$. We sum the joint probabilities over all possible combinations where $X = k$ and $Y = z - k$.
- The sum is $\sum_{k=1}^{z-1} \mathbb{P}(X = k)\mathbb{P}(Y = z - k) = \sum_{k=1}^{z-1} \alpha(1 - \alpha)^{k-1}\beta(1 - \beta)^{z-k-1}$.
- Rearranging the terms by pulling constants out of the sum yields: $\alpha\beta(1 - \beta)^{z-2} \sum_{k=1}^{z-1} \left(\frac{1 - \alpha}{1 - \beta}\right)^{k-1}$.
- The summation is a finite geometric series with common ratio $r = \frac{1 - \alpha}{1 - \beta}$ and $z - 1$ terms.
- Applying the geometric sum formula gives $\frac{1 - r^{z-1}}{1 - r} = \frac{1 - \left(\frac{1 - \alpha}{1 - \beta}\right)^{z-1}}{1 - \frac{1 - \alpha}{1 - \beta}}$.
- Multiplying the numerator and denominator by $(1 - \beta)^{z-1}$ gives $\frac{(1 - \beta)^{z-1} - (1 - \alpha)^{z-1}}{(1 - \beta) - (1 - \alpha)} = \frac{(1 - \beta)^{z-1} - (1 - \alpha)^{z-1}}{\alpha - \beta}$.
- Multiplying by the leading constant $\frac{\alpha\beta}{1 - \beta}$ simplifies the final result to $\frac{\alpha\beta}{\alpha - \beta} ((1 - \beta)^{z-1} - (1 - \alpha)^{z-1})$.

Solution to Q7

- Let Amit's time be $X \sim \text{Exp}(\frac{1}{5})$ and Supriya's time be $Y \sim \text{Exp}(\frac{1}{4})$.
- We need to compute the conditional probability $\mathbb{P}(X < Y \mid X > 1) = \frac{\mathbb{P}(1 < X < Y)}{\mathbb{P}(X > 1)}$.
- First, compute the denominator: $\mathbb{P}(X > 1) = \int_1^\infty \frac{1}{5}e^{-x/5}dx = e^{-1/5}$.
- Next, compute the numerator: $\mathbb{P}(1 < X < Y) = \int_1^\infty \left(\int_x^\infty \frac{1}{4}e^{-y/4}dy\right) \frac{1}{5}e^{-x/5}dx$.
- Evaluate the inner integral: $\int_x^\infty \frac{1}{4}e^{-y/4}dy = e^{-x/4}$.
- Substitute into the outer integral: $\int_1^\infty \frac{1}{5}e^{-x/5}e^{-x/4}dx = \int_1^\infty \frac{1}{5}e^{-9x/20}dx$.
- Evaluating this integral gives $\frac{1}{5} \left[-\frac{20}{9}e^{-9x/20}\right]_1^\infty = \frac{4}{9}e^{-9/20}$.
- Finally, form the ratio: $\frac{\frac{4}{9}e^{-0.45}}{e^{-0.20}} = \frac{4}{9}e^{-0.25}$.

Solution to Q8

- X and Y are independent uniform variables on $(0, 1)$, meaning their joint density is $f_{X,Y}(x, y) = 1$ for $x, y \in (0, 1)$.
- We transform to polar coordinates: $X = R \cos \Theta$ and $Y = R \sin \Theta$.
- **Part 1:** The Jacobian of the transformation from (R, Θ) to (X, Y) is R . Thus, the joint density is $f_{R,\Theta}(r, \theta) = f_{X,Y}(r \cos \theta, r \sin \theta) \cdot r = r$.
- The domain maps from the unit square $(0, 1) \times (0, 1)$. This restricts r based on θ .
- For $0 < \theta \leq \frac{\pi}{4}$, the boundary is $x = 1 \implies r \cos \theta = 1 \implies r = \sec \theta$.

- For $\frac{\pi}{4} < \theta < \frac{\pi}{2}$, the boundary is $y = 1 \implies r \sin \theta = 1 \implies r = \csc \theta$.
- Thus, $f_{R,\Theta}(r, \theta) = r$ on the region defined by $0 < \theta \leq \frac{\pi}{4}$, $0 < r < \sec \theta$ and $\frac{\pi}{4} < \theta < \frac{\pi}{2}$, $0 < r < \csc \theta$.
- **Part 2:** Marginal of Θ is found by integrating r : $f_{\Theta}(\theta) = \int_0^{\sec \theta} r dr = \frac{1}{2} \sec^2 \theta$ for $0 < \theta \leq \frac{\pi}{4}$, and $f_{\Theta}(\theta) = \int_0^{\csc \theta} r dr = \frac{1}{2} \csc^2 \theta$ for $\frac{\pi}{4} < \theta < \frac{\pi}{2}$.
- Marginal of R : For $0 < r \leq 1$, θ can range from 0 to $\frac{\pi}{2}$, so $f_R(r) = \int_0^{\pi/2} r d\theta = \frac{\pi}{2} r$. For $1 < r < \sqrt{2}$, θ is restricted between $\cos^{-1}(1/r)$ and $\sin^{-1}(1/r)$, giving $f_R(r) = \int_{\cos^{-1}(1/r)}^{\sin^{-1}(1/r)} r d\theta = r (\sin^{-1}(1/r) - \cos^{-1}(1/r))$.

Solution to Q9

- Since Z_i follows a standard normal distribution, we first find the moment generating function (MGF) of Z_i^2 .
- $M_{Z_i^2}(t) = \mathbb{E}[e^{tZ_i^2}] = \int_{-\infty}^{\infty} e^{tz^2} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$.
- Combine the exponents: $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}(1-2t)} dz$.
- This integral resembles a normal density with variance $(1-2t)^{-1}$. Solving gives $(1-2t)^{-1/2}$ for $t < \frac{1}{2}$.
- Because $Z_1, \dots, Z_n$ are independent, the MGF of $Y = \sum_{i=1}^n Z_i^2$ is the product of the individual MGFs.
- $M_Y(t) = \prod_{i=1}^n (1-2t)^{-1/2} = (1-2t)^{-n/2}$.
- This is exactly the MGF of a Gamma distribution with parameters $\alpha = \frac{n}{2}$ and $\lambda = \frac{1}{2}$.
- A Gamma distribution with these specific parameters is defined as a chi-square distribution with n degrees of freedom, concluding the proof.

Solution to Q10

- We are given $f_{(X,Y)}(x, y) = \frac{1}{2}(x+y)e^{-(x+y)}$ for $x, y \geq 0$.
- To find the density of $Z = X + Y$, we compute the convolution integral: $f_Z(z) = \int_0^z f_{(X,Y)}(x, z-x) dx$.
- Substitute the joint density: $f_Z(z) = \int_0^z \frac{1}{2}(x+z-x)e^{-(x+z-x)} dx = \int_0^z \frac{1}{2}ze^{-z} dx$.
- Since the integrand does not depend on x , this evaluates to $\frac{1}{2}ze^{-z}[x]_0^z = \frac{1}{2}z^2e^{-z}$ for $z \geq 0$.
- This is a Gamma distribution with shape parameter $\alpha = 3$ and rate parameter $\lambda = 1$.
- The mean of Z is calculated as $\mathbb{E}[Z] = \int_0^{\infty} z(\frac{1}{2}z^2e^{-z}) dz = \frac{1}{2} \int_0^{\infty} z^3e^{-z} dz$.
- Using the properties of the Gamma function ($\int_0^{\infty} t^{n-1}e^{-t} dt = (n-1)!$), we get $\frac{1}{2} \cdot 3! = \frac{6}{2} = 3$.

Solution to Q11

- Let $U = \min\{X, Y\}$ and $V = \max\{X, Y\}$. The sum $U + V = X + Y$, and the product $UV = XY$.
- The expectation of the sum is $\mathbb{E}[U + V] = \mathbb{E}[X] + \mathbb{E}[Y] = \frac{1}{2} + \frac{1}{2} = 1$.
- Since X and Y are independent, the expectation of the product is $\mathbb{E}[UV] = \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] = \frac{1}{4}$.
- To find $\mathbb{E}[U]$, note that $\mathbb{P}(U > u) = \mathbb{P}(X > u)\mathbb{P}(Y > u) = (1 - u)^2$. The density of U is the negative derivative, $2(1 - u)$.
- $\mathbb{E}[U] = \int_0^1 u \cdot 2(1 - u)du = 2 \left[\frac{u^2}{2} - \frac{u^3}{3} \right]_0^1 = 2 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3}$.
- Using $\mathbb{E}[U] + \mathbb{E}[V] = 1$, we deduce $\mathbb{E}[V] = \frac{2}{3}$.
- The covariance is $Cov(U, V) = \mathbb{E}[UV] - \mathbb{E}[U]\mathbb{E}[V] = \frac{1}{4} - \left(\frac{1}{3} \right) \left(\frac{2}{3} \right) = \frac{1}{4} - \frac{2}{9}$.
- Calculating the difference yields $\frac{9-8}{36} = \frac{1}{36}$.

Solution to Q12

- **Part i:** $X, Y \sim \text{Exp}(1)$.
- $X + Y$: The convolution of two independent $\text{Exp}(1)$ variables results in a $\text{Gamma}(2, 1)$ distribution with PDF ze^{-z} for $z > 0$.
- $X - Y$: This yields a Laplace distribution. The PDF is $\frac{1}{2}e^{-|z|}$ for $z \in \mathbb{R}$.
- X/Y : Let $W = X/Y$. Using the ratio distribution formula, $f(w) = \int_0^\infty ye^{-wy}e^{-y}dy = \int_0^\infty ye^{-y(w+1)}dy = \frac{1}{(w+1)^2}$ for $w > 0$.
- $\min\{X, Y\}$: The minimum of two $\text{Exp}(1)$ variables is $\text{Exp}(2)$ with PDF $2e^{-2z}$ for $z > 0$.
- $\max\{X, Y\}$: The CDF is $\mathbb{P}(X \leq z, Y \leq z) = (1 - e^{-z})^2$. Differentiating yields the PDF $2e^{-z}(1 - e^{-z})$ for $z > 0$.
- $\frac{X}{X+Y}$: Let $W = \frac{X}{X+Y}$ and $S = X + Y$. The Jacobian transformation leads to the PDF $f(w) = 1$ for $0 < w < 1$, which is a $\text{Uniform}(0, 1)$ distribution.
- **Part ii:** We analyze the transformation $U = X + Y, V = X - Y$. The inverse is $X = \frac{U+V}{2}, Y = \frac{U-V}{2}$, with Jacobian determinant $\frac{1}{2}$.
- The joint PDF is $f_{U,V}(u, v) = f_X\left(\frac{u+v}{2}\right)f_Y\left(\frac{u-v}{2}\right) \cdot \frac{1}{2} = \frac{1}{2}e^{-u}$ for $u > 0$ and $-u < v < u$.
- Since the marginal for U is ue^{-u} (from Part i), the conditional density is $f_{V|U}(v|u) = \frac{f(u,v)}{f_U(u)} = \frac{1}{2u}$ for $-u < v < u$.
- **Part iii:** Yes, $U = X + Y$ and $W = \frac{X}{X+Y}$ are independent. The joint PDF in the transformed coordinates factors perfectly into $f_U(u)f_W(w) = (ue^{-u}) \cdot 1$. Thus, they are independent.

Solution to Q13

- We start with the Law of Total Variance: $Var(Y) = \mathbb{E}[Var(Y|X)] + Var(\mathbb{E}[Y|X])$.
- Rearranging gives $\mathbb{E}[Var(Y|X)] = Var(Y) - Var(\mathbb{E}[Y|X])$.
- We know that the conditional expectation $\mathbb{E}[Y|X]$ is the minimum mean squared error estimator of Y given X .
- This implies it predicts Y at least as well as the best linear predictor $L(X) = aX + b$.
- Because X and Y have zero means, $L(X) = \rho \frac{\sigma_Y}{\sigma_X} X$. The variance of this linear predictor is $Var(L(X)) = \rho^2 \frac{Var(Y)}{Var(X)} Var(X) = \rho^2 Var(Y)$.
- Since $\mathbb{E}[Y|X]$ is an optimal predictor, its variance explains at least as much of $Var(Y)$ as the linear predictor does. Thus, $Var(\mathbb{E}[Y|X]) \geq \rho^2 Var(Y)$.
- Substituting this bound into our rearranged equation gives $\mathbb{E}[Var(Y|X)] \leq Var(Y) - \rho^2 Var(Y) = (1 - \rho^2) Var(Y)$.

Solution to Q14

- **Part i:** First, determine the normalizing constant. Integrate $x(y-x)e^{-y}$ over $0 \leq x \leq y < \infty$. Using the substitution $v = y - x$, the joint density factors into $xe^{-x}ve^{-v}$, establishing X and $Y - X$ as independent Gamma(2,1) variables. This gives $c = 1$.
- Marginal of X : $f_X(x) = \int_x^\infty x(y-x)e^{-y}dy = xe^{-x}$ for $x \geq 0$.
- Marginal of Y : $f_Y(y) = \int_0^y x(y-x)e^{-y}dx = e^{-y} \left[y\frac{x^2}{2} - \frac{x^3}{3} \right]_0^y = \frac{y^3}{6}e^{-y}$ for $y \geq 0$.
- The conditional distributions are: $f_{X|Y}(x|y) = \frac{x(y-x)e^{-y}}{(y^3/6)e^{-y}} = \frac{6x(y-x)}{y^3}$ for $0 \leq x \leq y$.
- $f_{Y|X}(y|x) = \frac{x(y-x)e^{-y}}{xe^{-x}} = (y-x)e^{-(y-x)}$ for $y \geq x$.
- **Part ii:** Calculate expectations using the conditional distributions.
- $\mathbb{E}[X|Y] = \int_0^y x \frac{6x(y-x)}{y^3} dx = \frac{6}{y^3} \left[y\frac{x^3}{3} - \frac{x^4}{4} \right]_0^y = \frac{6}{y^3} \left(\frac{y^4}{12} \right) = \frac{Y}{2}$.
- $\mathbb{E}[Y|X] = \int_x^\infty y(y-x)e^{-(y-x)} dy$. Substitute $v = y - x \implies y = v + x$. The integral becomes $\int_0^\infty (v+x)ve^{-v}dv = \int_0^\infty v^2e^{-v}dv + x \int_0^\infty ve^{-v}dv = 2 + X$.

Solution to Q15

- **Part a:** Compute the covariance between $U = X + Y$ and $V = X - Y$.
- $Cov(U, V) = Cov(X + Y, X - Y)$. Using bilinearity of covariance, this expands to $Cov(X, X) - Cov(X, Y) + Cov(Y, X) - Cov(Y, Y)$.
- Since X and Y are independent, $Cov(X, Y) = Cov(Y, X) = 0$.
- Since X and Y are identically distributed, $Cov(X, X) = Var(X)$ is equal to $Cov(Y, Y) = Var(Y)$.

- Thus, $Cov(U, V) = Var(X) - Var(Y) = 0$. This shows they are uncorrelated. They are not necessarily independent for arbitrary distributions.
- **Part b:** If X and Y follow a normal distribution $\mathcal{N}(0, 1)$, any linear combination of them forms a multivariate normal distribution.
- For variables that are jointly normal, a covariance of zero implies they are independent. Thus, in this specific case, U and V are independent.

Solution to Q16

- Given joint density $f_{X,Y}(x, y) = 2e^{-(x+y)}$ for $0 < x < y < \infty$.
- Marginal for X : Integrate over y . $f_X(x) = \int_x^\infty 2e^{-(x+y)} dy = 2e^{-x}[-e^{-y}]_x^\infty = 2e^{-2x}$ for $x > 0$.
- Marginal for Y : Integrate over x . $f_Y(y) = \int_0^y 2e^{-(x+y)} dx = 2e^{-y}[-e^{-x}]_0^y = 2e^{-y}(1 - e^{-y})$ for $y > 0$.
- Compute expectations: $\mathbb{E}[X] = \frac{1}{2}$ (from $\text{Exp}(2)$ distribution). $\mathbb{E}[Y] = \int_0^\infty y(2e^{-y} - 2e^{-2y}) dy = 2(1) - 2(\frac{1}{4}) = \frac{3}{2}$.
- Compute $\mathbb{E}[XY] = \int_0^\infty \int_x^\infty xy 2e^{-(x+y)} dy dx$. The inner integral evaluates to $xe^{-x}(x+1)$. The outer integral evaluates to $\int_0^\infty 2xe^{-2x}(x+1) dx = \int_0^\infty (2x^2 + 2x)e^{-2x} dx$.
- Using $\int_0^\infty t^n e^{-at} dt = \frac{n!}{a^{n+1}}$, we get $\frac{2(2)}{8} + \frac{2(1)}{4} = \frac{1}{2} + \frac{1}{2} = 1$.
- Covariance: $Cov(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 1 - (\frac{1}{2})(\frac{3}{2}) = 1 - \frac{3}{4} = \frac{1}{4}$.

Solution to Q17

- **Part a:** Define $W = \frac{Y}{X}$. If $\Theta = \tan^{-1}(W)$ is uniformly distributed on $(-\frac{\pi}{2}, \frac{\pi}{2})$, then by transformation, W must follow a standard Cauchy distribution.
- The probability density function of a standard Cauchy is given by $f_W(w) = \frac{1}{\pi(1+w^2)}$.
- Using the formula for the ratio of two independent variables, the PDF of $W = \frac{Y}{X}$ is $\int_{-\infty}^\infty |x|f(x)f(xw)dx$.
- Setting this equal to the Cauchy density yields the condition: $\int_{-\infty}^\infty f(x)f(xw)|x|dx = \frac{1}{\pi(1+w^2)}$ for all $w \in \mathbb{R}$.
- **Part b:** The standard normal density $f(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ satisfies this identity, as the ratio of two independent standard normal variables is exactly a standard Cauchy distribution.

Solution to Q18

- Given $f(x, y) = x + y$ on $(0, 1) \times (0, 1)$.
- Marginal density of X : $f_X(x) = \int_0^1 (x + y) dy = \left[xy + \frac{y^2}{2}\right]_0^1 = x + \frac{1}{2}$. By symmetry, $f_Y(y) = y + \frac{1}{2}$.

- $\mathbb{E}[X] = \int_0^1 x(x + \frac{1}{2})dx = \left[\frac{x^3}{3} + \frac{x^2}{4}\right]_0^1 = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$. By symmetry, $\mathbb{E}[Y] = \frac{7}{12}$.
- $\mathbb{E}[X^2] = \int_0^1 x^2(x + \frac{1}{2})dx = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$.
- $Var(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{5}{12} - \frac{49}{144} = \frac{60-49}{144} = \frac{11}{144}$. $Var(Y) = \frac{11}{144}$.
- $\mathbb{E}[XY] = \int_0^1 \int_0^1 xy(x+y)dydx = \int_0^1 x\left(\frac{x}{2} + \frac{1}{3}\right)dx = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$.
- $Cov(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \frac{1}{3} - \frac{49}{144} = \frac{48-49}{144} = -\frac{1}{144}$.
- Correlation coefficient $\rho = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} = \frac{-1/144}{11/144} = -\frac{1}{11}$.

Solution to Q19

- We are given $\rho_{(X,Y)} = \frac{1}{2}$, $Var(X) = 1$, and $Var(Y) = 4$.
- First, compute the covariance: $Cov(X, Y) = \rho\sqrt{Var(X)Var(Y)} = \frac{1}{2}\sqrt{1 \cdot 4} = 1$.
- We need to compute $Var(X - 3Y)$. Using the properties of variance, $Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2abCov(X, Y)$.
- Substituting $a = 1$ and $b = -3$: $Var(X - 3Y) = (1)^2Var(X) + (-3)^2Var(Y) + 2(1)(-3)Cov(X, Y)$.
- This simplifies to $Var(X) + 9Var(Y) - 6Cov(X, Y)$.
- Plug in the known values: $1 + 9(4) - 6(1) = 1 + 36 - 6 = 31$.